PLANES, PRESCRIPTION & COVERAGE

HOW THE TECHNOLOGIST DEFINES WHAT TO IMAGE

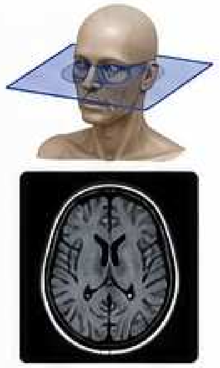
Proper plane orientation and prescription ensure the anatomy and clinical area of interest are fully covered with the right resolution and thickness. This is the foundation of every good MRI exam.

KEY CONCEPT

The technologist chooses the plane, field of view (FOV), slice thickness, and number of slices to match the clinical question while balancing resolution, SNR, and scan time.

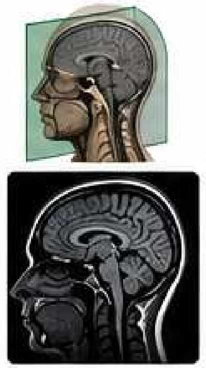
1 ANATOMICAL PLANES

AXIAL
Transverse



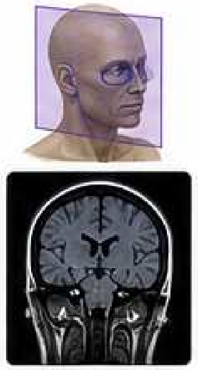
Slices run left ↔ right.
View from feet looking up.

SAGITTAL
Side to side



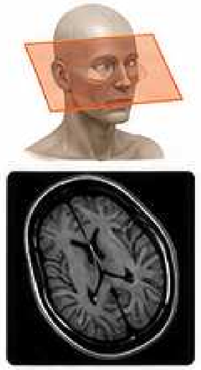
Slices run right ↔ left.
View from patient's right.

CORONAL
Front to back



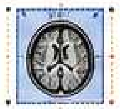
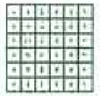
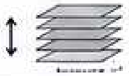
Slices run anterior ↔ posterior.
View from front.

OBLIQUE
Angled to anatomy



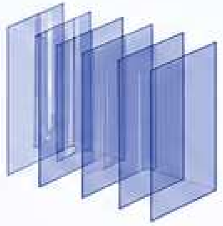
Angled to follow anatomy
(e.g., orbit, spine, inner ear).

2 PRESCRIPTION ELEMENTS

FOV (Field of View)	The size of the area being imaged.		• Smaller FOV = higher resolution • Larger FOV = more coverage
MATRIX (Phase × Frequency)	Number of data points in each direction.		• Higher matrix = smaller pixel size (better detail) • Lower matrix = larger pixels (faster scan, lower detail)
SLICE THICKNESS	The thickness of each slice.		• Thinner = better detail, lower SNR • Thicker = higher SNR, more partial volume
# OF SLICES (OR COVERAGE)	Total number of slices to cover the anatomy.		• More slices = more coverage • Depends on thickness and spacing
SLICE GAP / SPACING	Space between slices.		• 0 mm = contiguous • Gap = may miss small findings

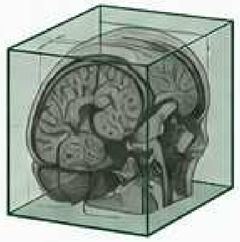
3 2D vs 3D ACQUISITION

2D (MULTI-SLICE)



- Each slice excited separately
- Thinner slices achievable
- Longer scan time
- Less flexible reformatting

3D (VOLUME)



- Whole volume excited
- Slightly longer scan
- Thin slices can be reformatted in any plane
- Excellent for isotropic imaging

4 COVERAGE EXAMPLES (BRAIN)

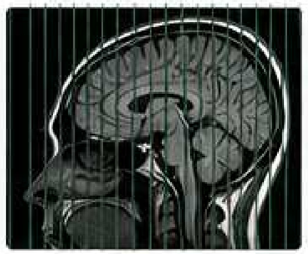
AXIAL COVERAGE

From foramen magnum through the vertex.

CORONAL COVERAGE

From ear to ear (or wider as needed).

SAGITTAL COVERAGE



Midline to lateral brain (or wider).

5 HOW CHOICES AFFECT THE IMAGE

Change This	If You...	Result	Effect on Image	Effect on Scan Time
Decrease FOV	Image a smaller area	Higher resolution (smaller pixels)	More detail of the area of interest	Usually no change (or slightly shorter)
Increase FOV	Need more coverage	Lower resolution (larger pixels)	Less detail	Usually no change (or slightly shorter)
Increase Matrix	Increase resolution	Smaller pixels	Sharper / more detail (higher spatial resolution)	Longer scan
Decrease Matrix	Need faster scan	Larger pixels	Less detail	Shorter scan
Decrease Slice Thickness	Need more detail	Thinner slices	Better depiction of small structures	Longer scan, lower SNR
Increase Slice Thickness	Need higher SNR or faster scan	Thicker slices	Higher SNR, less partial volume	Shorter scan
Increase # of Slices	Cover more anatomy	More slices	More coverage	Longer scan
Increase Gap	Speed up scan	Less slices needed	Risk of missing anatomy between slices	Shorter scan

ENGINEER VIEW: WHAT TO CHECK WHEN IMAGE QUALITY IS POOR

COVERAGE
- Is the anatomy fully covered?
- Are landmarks correct?

FOV
- Too small? → Aliasing/wrap
- Too large? → Poor resolution

MATRIX
- Too low? → Blocky, less detail
- Too high? → Long scan, may reduce SNR

SLICE THICKNESS
- Too thick? → Partial volume
- Too thin? → Low SNR, long scan

GAP / SPACING
- Too large? → Gaps in anatomy
- 0 mm preferred for most studies

2D vs 3D
- Need thin slices in any plane? Use 3D.

BEST PRACTICE

Always start troubleshooting by asking:
"Is the coverage correct and are the parameters appropriate for the clinical question?"

Good prescriptions = good images. Understand the choices the technologist makes so you can recognize when an issue is a parameter choice — or a true system problem.